

Reliability of J-PARC accelerator system over the past decade

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The Japan Proton Accelerator Research Complex (J-PARC) is a multipurpose facility for scientific experiments. The accelerator complex consists of a 400-MeV linac, a 3-GeV Rapid-Cycling Synchrotron (RCS) and a 30-GeV Main Ring synchrotron (MR). The RCS delivers a proton beam to the neutron target and MR, and the MR delivers the beams to the neutrino target and the Hadron Experimental Facility. The accelerator facilities in J-PARC have been commissioned since January 2007. Although there were some accidental suspensions in the past decade, such as the Great East Japan Earthquake in March of 2011 and recovery work from it, the recent availability of the facilities reaches 90% or more for the neutron users and around 80% for MR users. This is the result of our continued efforts for the improvement and user operation. This report summarizes the major causes of suspension, and analyzes the reliability statistics of the J-PARC accelerator system. As an example of our efforts, we also describe how we could recover the Mean Time Between Failure (MTBF) longer than a few hours, which dropped down to less than one hour.

KEYWORDS: J-PARC accelerator, Reliability, Availability, Statistics

1. Introduction

The Japan Proton Accelerator Research Complex (J-PARC) is a multipurpose facility for conducting scientific experiments. The accelerator complex consists of a 400 MeV linac, a 3 GeV rapid-cycling synchrotron (RCS), and a 30 GeV main ring synchrotron (MR). The beam commissioning of the linac was started in November of 2006 [1], and the RCS began delivering a proton beam to the Material and Life science experimental Facility (MLF) and the MR in May of 2008 [2]. User operation of the MLF began in December of 2008 [3]. The MR has two extraction lines: a slow extraction (SX) line that delivers a proton beam to the Hadron Experimental Facility, and a fast extraction (FX) line that delivers the beam to the neutrino target. In January of 2009, we achieved a slow beam extraction for the hadron beamline [4], and the neutrino beamline commissioning began in April of 2009. The regular neutrino experiments started in January of 2010 [5]. Following this, we continued the operation but encountered some accidental suspensions. This report summarizes the major causes of suspension, and analyzes the reliability statistics of the J-PARC accelerator system.